Null and Alternative Hypotheses

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Inferential statistics; Sampling distribution and standard error.
- **Leads to:** Z-test and rejection regions; Type I and II errors, power, tails; P-values; T-tests: one-sample, two-sample, paired; One-sample proportion test; Chi-square tests; One-way ANOVA.
- **Compare with:** Confidence intervals.

1. Overview

Key point: A hypothesis test turns an idea into two opposite statements, the null and the **alternative hypothesis** (G-193), and asks whether a sample gives strong enough evidence against the null.

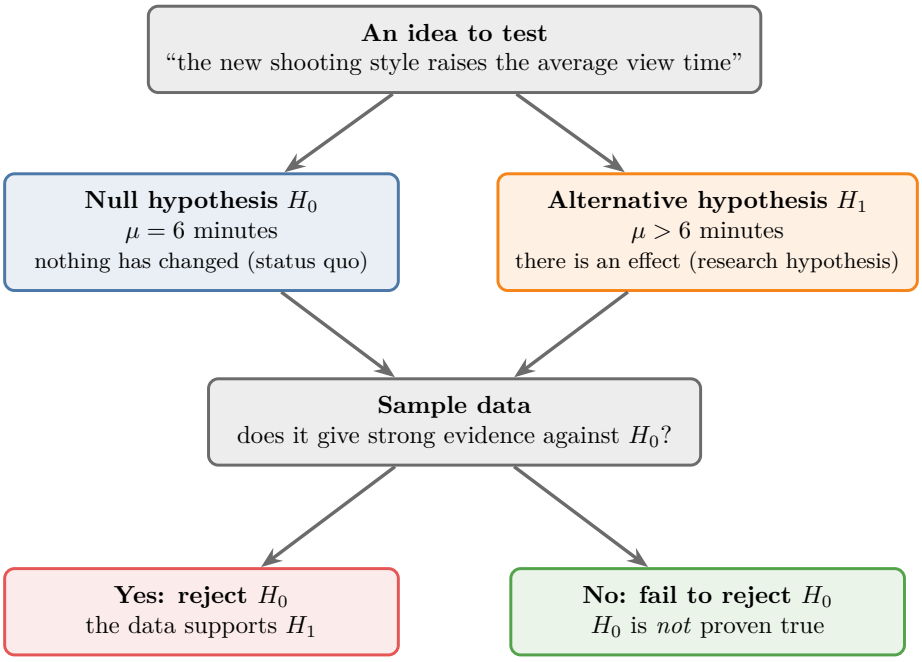


Figure 1: The idea behind every hypothesis test

Hypothesis testing (G-913) checks a claim about a population parameter with a sample (see population and sample). Figure 1 shows the whole idea on one example: we write the claim as two statements, look at the data, and either reject the first statement or fail to reject it.

This Note covers:

- the problem hypothesis testing solves;
- the **null hypothesis** (G-1361) and the alternative hypothesis, and how to write them;
- three rules for using them, with the courtroom analogy;
- the eight steps of a hypothesis test.

The next Notes run these steps on real numbers: the logic of a test with a **z-test** (G-2143), and the definition of a p-value with the method used in practice.

2. The problem hypothesis testing solves

> **Key point:** One result, or a few, can be good or bad by chance; **hypothesis testing** (G-913) tells us whether a sample is strong enough evidence for a claim about the whole population.

2.1 An online channel tries a new style

Suppose an online channel records its lessons on a digital writing pad. Its analytics show that, over all its recorded lessons, people watch for **6 minutes** on average. Successful channels seem to film in front of a whiteboard instead, so the owner has an idea: the whiteboard style would raise the average view duration.

The obvious test is to try it. The first whiteboard lesson is watched for 30 minutes on average. Can we conclude that the new style works?

No. One lesson proves very little:

- its topic may have been popular (say, a lesson on ChatGPT that the whole world wants to watch);
- the presenter may have looked especially good that day;
- any other random reason may have lifted that one lesson.

So the channel records five more lessons in the new style. Their average view durations are 7, 9, 5, 11 and 13 minutes. Their mean, 9 minutes, is above 6. Is that enough? The question has not changed: are five lessons enough evidence, or could the old style give such numbers by chance?

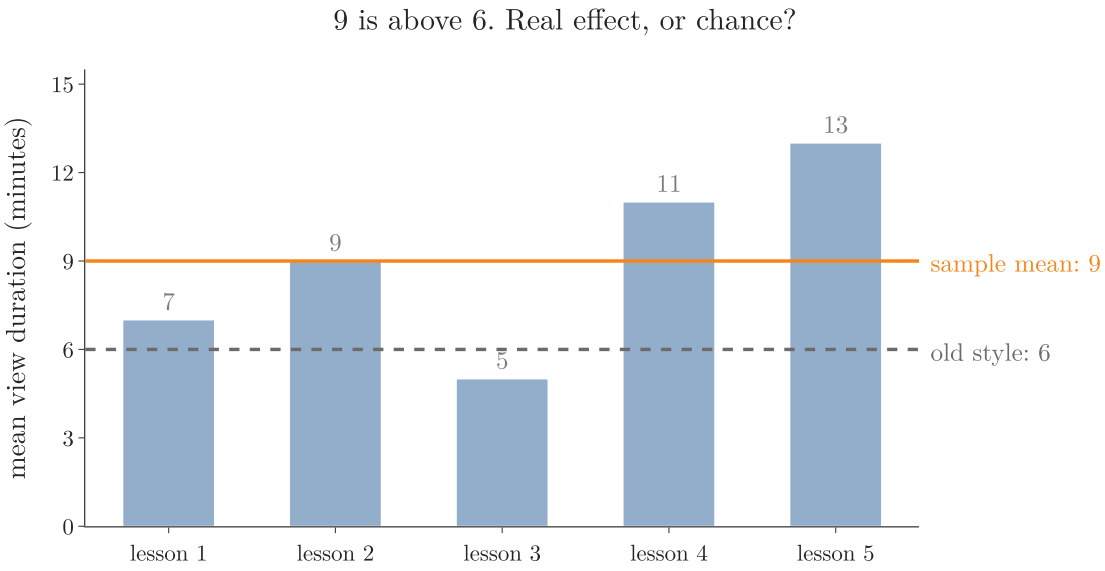


Figure 2: The five new-style lessons. Their mean, 9 minutes, sits above the old 6-minute average, but two lessons are close to 6 and one is below it

Figure 2 shows the five lessons against the old average. The spread is the problem: with lessons ranging from 5 to 13 minutes, a mean of 9 might come from the old style by chance. The Extra at the end of Section 6 gives the verdict.

2.2 The same question in business

A chips company sells a green packet and launches a purple one. Before replacing green with purple everywhere, it wants to know whether the purple packet really sells better. It can ask customers for feedback, but then: how many customers, and how good must the feedback be before we believe it?

A small channel can afford to try and see. A company such as Amazon or Flipkart, where one wrong decision costs a lot of money, needs a method. That method is **hypothesis testing** (G-913).

2.3 Definition

A **statistical hypothesis test** (G-1881) is a method of statistical inference used to decide whether the data at hand sufficiently supports a particular hypothesis. **Hypothesis testing** (G-913) lets us make probabilistic statements about population parameters.

In the channel example the population parameter is μ, the mean view duration of all lessons the channel could record in the new style. The new lessons are a sample. Hypothesis testing tells us what the sample can say about μ.

Hypothesis testing is **inferential statistics** (G-944): reasoning from a sample to a population (see descriptive and inferential statistics). Inferential statistics is used constantly in business,

finance and economics.

3. The null hypothesis

> **Key point:** The null hypothesis H_0 says that nothing has changed: no effect, no difference, no relationship.

Every hypothesis test starts with two statements. The first is the **null hypothesis** (G-1361), written H_0:

> a statement that assumes there is no significant effect or relationship between the variables being studied. It is the starting point of the test and stands for the **status quo** (G-1887), "no effect until proven otherwise".

Two examples:

- **The channel.** We want to show that the new style raises the view duration. The null hypothesis says the opposite: the old and the new style make no difference, and the mean view duration is still 6 minutes.
- **A chips packet.** A packet says "100 g" on the back, and we suspect the true mean weight is different, either more or less. The null hypothesis says the mean weight of the packets is exactly 100 g.

In both cases H_0 is the boring statement: whatever was true before is still true. The test then tries to prove it wrong.

Why test "no change" and not the change we hope for? The five lessons average 9 minutes, so we could test the statement "the new style adds 3 minutes". But another five lessons might average 8.5 or 9.4, and "adds 2.5 minutes" and "adds 3.4 minutes" would then be just as reasonable. There is no end to such statements, and we cannot even write one down before we have data. "No change" is different: it is one statement, it is fixed before any data arrives, and its value is known exactly (the old mean, 6). So every test checks that one statement.

H_0 does not claim that every new lesson scores exactly 6. It claims that the mean is still 6 and that any difference in a sample is chance. Figure 3 draws what chance alone would do. If H_0 were true, the mean of 5 lessons would still wander around 6, because each lesson varies; with the spread of our five lessons, the one-sample t-test (see the t statistic step by step) says that 5.1 percent of such means would land at 9 or beyond. The test asks whether our result is too far out to blame on chance.

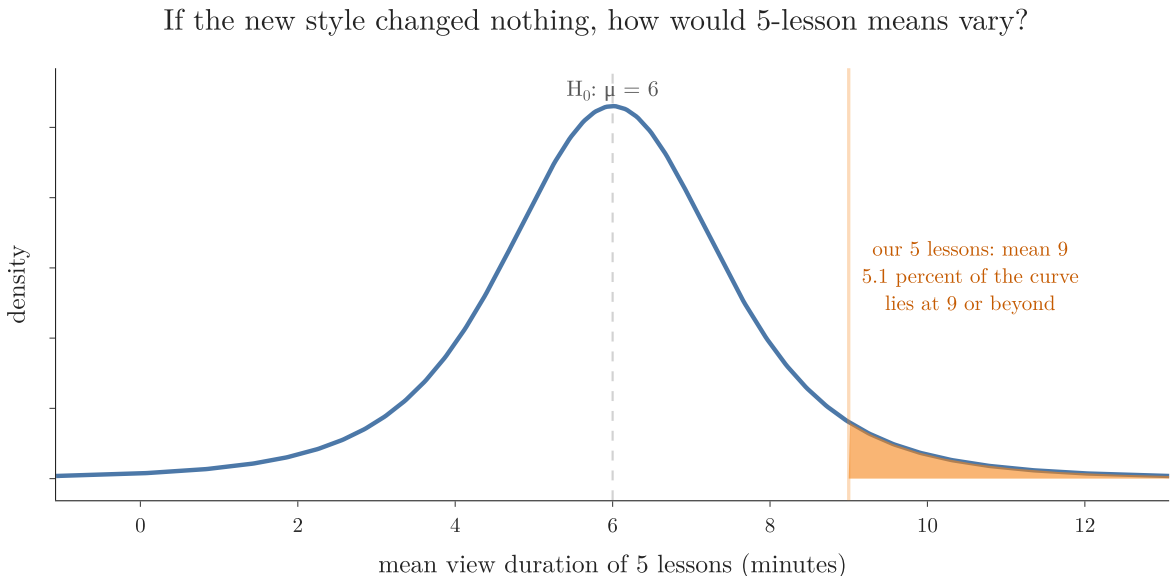


Figure 3: If H0 is true (mean 6), the means of 5 lessons vary by chance around 6; the curve is the one-sample t-test's picture of that variation, using the five lessons' spread (s = 3.16). 5.1 percent of it lies at the observed mean 9 or beyond.

4. The alternative hypothesis

> **Key point:** The alternative hypothesis H_1 contradicts H_0 and claims there is an effect; the two can never both be true.

The second statement is the **alternative hypothesis** (G-193), written H_1 or H_a:

> a statement that contradicts the **null hypothesis** (G-1361) and claims there is a significant effect or relationship between the variables being studied.

The two statements are **mutually exclusive** (G-1286): they cannot both be true, and the test ends by siding with one of them. They need not cover every case: for the channel, a mean below 6 fits neither statement (the grey part of the line in Figure 4). For our examples:

Example	H_0 (no effect)	H_1 (an effect)
Channel: mean view duration in minutes	$\mu = 6$	$\mu > 6$
Chips: mean packet weight in grams	$\mu = 100$	$\mu \neq 100$

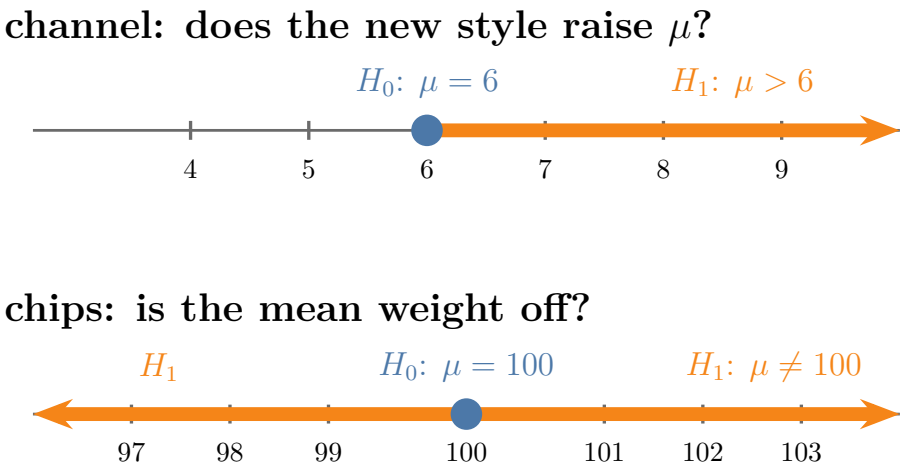


Figure 4: The two examples on a number line. H_0 is a single value (blue dot); H_1 covers the values it rules out (orange): one side for the channel, both sides for the chips

Figure 4 draws the table: the channel's H_1 points one way, the chips' H_1 points both ways.

Figure 5 shows what the test will weigh. Each lesson is a dot, and each bar is its distance to a horizontal line. The line starts at 6, the mean that H_0 claims, and moves to 9, the mean of the lessons themselves.

- **Our five lessons (left).** The distances to 6 add up to 17 minutes:

  $$1 + 3 + 1 + 5 + 7 = 17$$

  The distances to 9 add up to 12:

  $$2 + 0 + 4 + 2 + 4 = 12$$

  The line at 9 fits only a little better than the line at 6, because the lessons are spread out from 5 to 13. So H_0 is hard to reject: if H_0 were true, chance alone would give a mean this far above 6 in 5.1 percent of such samples (Figure 3).
- **Five steadier lessons (right).** Suppose the lessons had been 8, 9, 8, 10 and 10 minutes: the same mean 9, but close together. The total distance falls from 15 to 4. The line at 6 fits these lessons far worse than their own mean, and the same one-sample t-test says chance alone would do this in only 0.1 percent of such samples: strong evidence against H_0.

The same mean of 9 is weak evidence in one case and strong evidence in the other. A test compares how far the sample is from H_0 with how much the sample varies by itself.

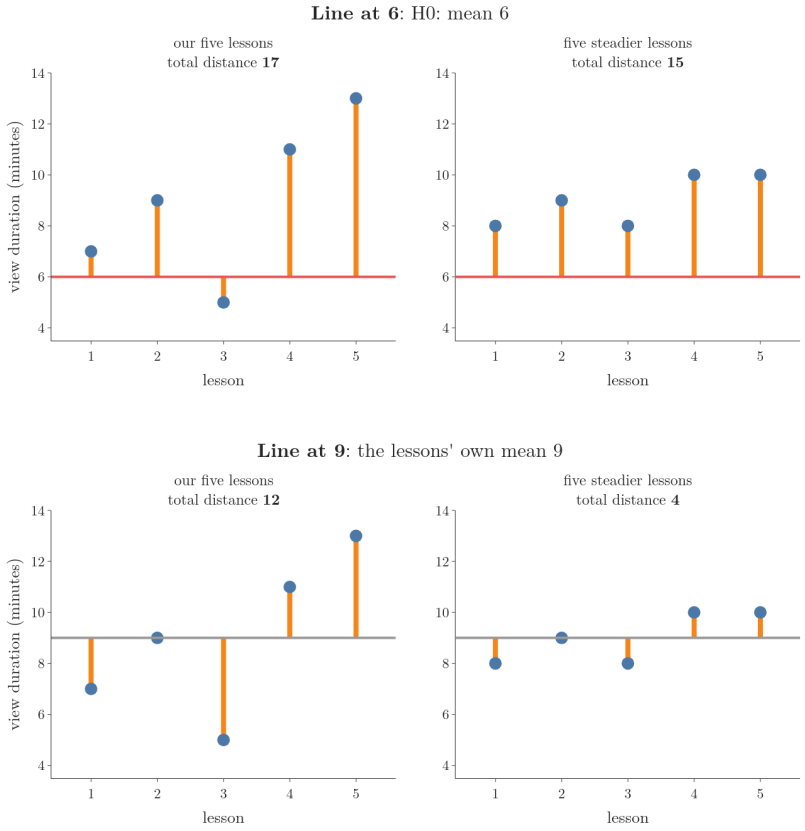


Figure 5: Five lessons and their distances (orange) to a line that moves from 6, the mean under H_0, to 9, the lessons' own mean. Left: our lessons, total distance 17 to 12. Right: five steadier lessons with the same mean, 15 to 4. Adapted from StatQuest, "Alternative Hypotheses: Main Ideas!!!", which compares distances to one mean and to two group means.

The channel's H_1 has a direction: the new style should **increase** the duration. The chips' H_1 has none: the weight is wrong, whether too high or too low. This difference decides between a **one-tailed test** (G-1385) and a two-tailed test (see the one-tailed and two-tailed tests).

Two other names are common in books and exam questions:

- H_0 is called the **status quo** (G-1887): the current state of things;
- H_1 is called the **research hypothesis** (G-1677): the idea that comes out of research. The channel owner studied other channels and concluded that the whiteboard style might help.

> **Extra:** The hypotheses are always about the **population** parameter (μ), never about the sample mean ($\bar{x}$). We already know $\bar{x} = 9$ for the five new lessons; there is nothing to test about it. The open question is μ, the mean of all future lessons. Writing $H_0 : \bar{x} = 6$ is a common slip.

5. Three rules for H_0 and H_1

> **Key point:** H_0 says nothing new is happening; the test collects evidence against H_0; and failing to reject H_0 does not prove it true.

5.1 How to tell which statement is H_0

Faced with a new problem, we may wonder which statement should be H_0. The rule of thumb: H_0 **is the statement that nothing new is happening.**

- We changed the filming style, and the mean view duration is still 6 minutes: H_0.
- We opened packets and weighed them, and the mean is still 100 g: H_0.

The statement with the change, the difference or the effect is H_1.

> **Extra:** The equals sign always goes in H_0. The test needs one exact value to compute with ($\mu = 6$), and only H_0 provides it.

5.2 The goal: evidence against H_0

A hypothesis test does not try to prove H_1 directly. The test assumes H_0 and then uses the data to collect **evidence against** H_0. If the evidence is strong enough, we reject H_0, and H_1 is what remains.

5.3 Failing to reject H_0 does not prove it

The most important rule: **failing to reject the null hypothesis does not mean the null hypothesis is true.** Failing to reject only means the data did not bring enough evidence against it.

A courtroom makes this clear:

Court	Hypothesis test
The accused is presumed innocent	H_0 is assumed true: no crime
The prosecutor claims the accused is guilty	H_1: there was a crime
The prosecutor brings evidence	We collect sample data
Strong evidence: verdict “guilty”	Reject H_0
Weak evidence: verdict “not guilty”	Fail to reject H_0

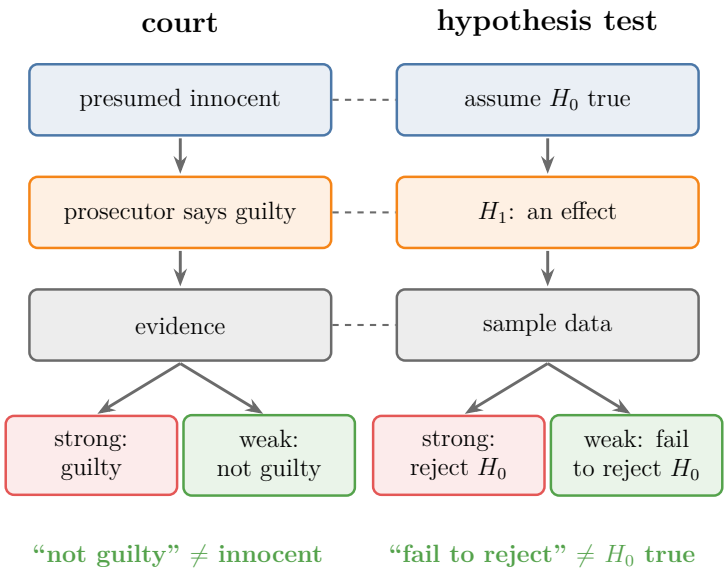


Figure 6: The courtroom and the hypothesis test, step for step. Both end in one of two verdicts, and the weak-evidence verdict proves nothing

Figure 6 lines up the two procedures. Watch the green boxes: neither "not guilty" nor "fail to reject" is a proof.

A "not guilty" verdict does not prove that no crime happened. The verdict says the prosecutor could not prove it. In the same way, if the five lessons do not let us reject H_0, the new style may still work: our evidence was simply too weak to show it.

For the same reason, rejecting H_0 for the channel would not show that the whiteboard style is the best possible. Rejecting H_0 would only show that the new style beats the old average of 6 minutes.

> **Extra:** The courtroom logic is why we say "fail to reject H_0" and avoid "accept H_0". A small sample gives a test little power to detect a real effect: in the power of a test the same test detects a true mean of 52 with probability 0.71 for 30 employees and 0.99 for 100. "Accepting" H_0 after such a test would turn weak evidence into a false certainty. Altman and Bland (1995) make the same point: a non-significant result does not show that there is no effect.

6. The eight steps of a hypothesis test

> **Key point:** State H_0 and H_1, choose α, check assumptions, choose the test and its statistic, compute it, decide, and interpret.

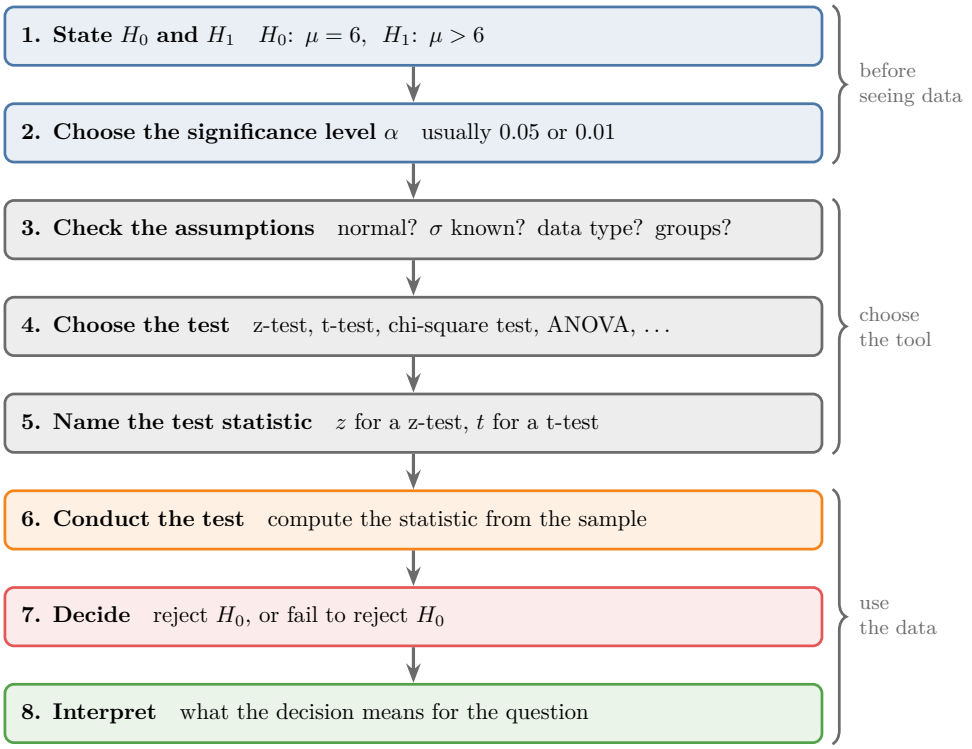


Figure 7: The eight steps of a hypothesis test (rejection region approach)

There are two ways to carry out a test. The **rejection region approach** (G-1663) compares the test statistic with a fixed boundary (see the rejection region and the critical value). The **p-value approach** (G-1431) computes one extra number that also measures how strong the evidence is (see the definition of a p-value); it is the approach used in practice. Both follow the steps in Figure 7.

1. **State H_0 and H_1.** For the channel: $H_0 : \mu = 6$, $H_1 : \mu > 6$.
2. **Choose a significance level** α (G-1801). Usually 0.05 (5%), sometimes 0.01 (1%). The significance level is the probability of rejecting H_0 when H_0 is actually true: at 5%, about 5 tests in 100 with a true H_0 would wrongly reject it. How rare is too rare explains it fully.
3. **Check the assumptions about the data.** Is the data normally distributed? Do we know the population standard deviation σ? Is the data numerical or categorical? Do we have one **feature** (G-772; one measured variable, one column of the data table) or several, one group or several?
4. **Choose the test.** The assumptions decide it. For example:
   - normal data (or a large sample) and σ known: the one-sample **z-test** (G-2143);
   - σ unknown: the **t-test** (G-1940; see from the z-test to the t-test);
   - categorical data: the **chi-square test** (G-381);
   - several group means: **ANOVA** (G-203).
5. **Name the test statistic.** The **test statistic** (G-1963) is the number the test computes from the sample. A z-test computes a z-score, called the z statistic; a t-test computes

the t statistic (as in the t-procedure formula).

6. **Conduct the test.** Compute the test statistic from the sample.
7. **Decide.** Based on the statistic, reject H_0 or fail to reject it.
8. **Interpret the result.** Translate the decision back into the real question. If the channel rejected H_0, the interpretation would be: "filming in the new style increases the mean view duration".

Steps 1 and 2 come before looking at the data: the decision needs a fixed boundary, and the boundary comes from α and the direction of H_1. Choosing them after seeing the results changes the error rate. For example, if we picked the direction of H_1 after seeing which side of 6 the sample mean fell on, a true H_0 would be rejected whenever $|z| > 1.645$ (here z is the z statistic: how many standard errors the sample mean lies from 6; $|z|$ drops its sign). That has this probability:

$$2 \times 0.05 = 0.10$$

This is double the α we claimed.

> **Extra:** Are the five lessons (7, 9, 5, 11, 13 minutes) enough evidence that $\mu > 6$? Back to the five lessons answers it: at the 5% level, just barely not.

7. Summary

	Null hypothesis H_0	Alternative hypothesis H_1
Says	no effect, no difference	there is an effect or difference
Other name	status quo	research hypothesis
Contains	the equals sign ($\mu = 6$)	$>$, $<$ or $\neq$
Role in the test	assumed true; we look for evidence against it	what remains if H_0 is rejected
Channel example	$\mu = 6$ minutes	$\mu > 6$ minutes
Chips example	$\mu = 100$ g	$\mu \neq 100$ g

- The equals sign goes in H_0 because the test needs one exact value to compute with ($\mu = 6$), and only H_0 provides it.
- A single result, or a few, can be lucky; a hypothesis test asks whether a sample is strong enough evidence about a population parameter, so a costly decision (replace the green chips packet?) does not rest on chance.
- H_0 says nothing new is happening; H_1 contradicts it. They cannot both be true, so the test ends by siding with one of them.
- The test collects evidence against H_0, because "no change" is one fixed statement we can compute with, while the changes we hope for are endless. The two outcomes are "reject H_0" and "fail to reject H_0".
- Failing to reject H_0 does not prove H_0: "not guilty" is not "innocent", because a small sample may simply be too weak to show a real effect; so say "fail to reject", never "accept".
- Eight steps: hypotheses, α, assumptions, test, statistic, compute, decide, interpret, so the assumptions about the data decide which test is used.
- This answers the opening question: a test turns an idea into H_0 and H_1 and asks whether the sample is strong enough evidence against H_0.

8. Sources

Built from

- CampusX, "Session 45 - Hypothesis Testing Part 1 | DSMP 2023", YouTube, https://www.youtube.com/watch?v=S94mx6OL7kM
- StatQuest with Josh Starmer, "Hypothesis Testing and The Null Hypothesis, Clearly Explained!!!", YouTube, https://www.youtube.com/watch?v=0oc49DyA3hU
- StatQuest with Josh Starmer, "Alternative Hypotheses: Main Ideas!!!", YouTube, https://www.youtube.com/watch?v=5koKb5B_YWo

Other references

- Altman, D. G. and Bland, J. M. (1995). "Absence of evidence is not evidence of absence." *BMJ* 311(7003), 485.

9. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Alternative hypothesis (H_1, H_a) (G-193)	The claim of an effect, difference or relationship: the statement that contradicts the null hypothesis H_0.
Hypothesis testing (G-913)	Checking a claim about a population parameter with a sample.
Null hypothesis (H_0) (G-1361)	The statement of no effect, no difference or no relationship; assumed true until the data gives strong evidence against it.
Statistical hypothesis test (G-1881)	A method that uses sample data to decide whether a claim about a whole population (a hypothesis about a population parameter) is supported strongly enough; it is a form of statistical inference.
Status quo (G-1887)	Another name for the null hypothesis H_0: the current state of things, which the test keeps unless the data gives strong evidence against it.
Research hypothesis (G-1677)	Another name for the alternative hypothesis: the idea that came out of research.
P-value approach (G-1431)	Carrying out a test by computing a p-value, which also measures the strength of the evidence.
Test statistic (G-1963)	The number a test computes from the sample to make its decision, such as z or t.
Fail to reject H_0 (G-746)	The decision that the evidence against the null hypothesis H_0 is not strong enough to drop it; it does not prove H_0.
Reject H_0 (G-1661)	The decision that the data gives strong enough evidence against H_0.
Z-test (one-sample) (G-2143)	A test of whether a population's mean equals a claimed value μ_0, used when the population spread σ is known and $\bar{X}$ is normal: $z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$.
Inferential statistics (G-944)	Statistics that draws conclusions about a population from a sample.
Mutually exclusive events (G-1286)	Events that cannot happen at the same time; their intersection has probability 0.
One-tailed test (one-sided test) (G-1385)	A test whose H_1 has a direction ($>$ or $<$), with the whole rejection region in one tail.

Term	Meaning
Rejection region approach (G-1663)	Carrying out a test by checking whether the test statistic falls beyond a fixed boundary.
Significance level (α) (G-1801)	The chance, fixed before the test, of wrongly declaring an effect when there is none (rejecting H_0 when it is actually true); usually 0.05.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
T-test (G-1940)	A hypothesis test about means that uses the sample standard deviation and Student's t-distribution.
Chi-square test (G-381)	A test of whether categorical counts match expected counts (goodness of fit) or whether two categorical columns are related (independence).
ANOVA (G-203)	A test of whether several groups have the same mean (analysis of variance).